Assignment – 8.5

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Batch: 23

Task Description #1 (Username Validator – Apply AI in Authentication Context)

- Task: Use AI to generate at least 3 assert test cases for a function is_valid_username(username) and then implement the function using Test-Driven Development principles.
- Requirements:
  - Username length must be between 5 and 15 characters.
  - Must contain only alphabets and digits.
  - Must not start with a digit.
  - No spaces allowed.

Example Assert Test Cases:
assert is_valid_username("User123") == True assert is_valid_username("12User") == False assert is_valid_username("Us er") == False Expected Output #1:

- Username validation logic successfully passing all AI-generated test cases

Code:

```python
def is_valid_username(username):
    '''
    checks whether a username is valid.

    conditions:
    - length should be between 5 and 15 characters
    - only letters and numbers are allowed
    - should not start with a number
    - spaces are not allowed
    '''

    if len(username) < 5 or len(username) > 15:
        return False
    if not username.isalnum():
        return False
    if username[0].isdigit():
        return False
    return True

# assert test cases
assert is_valid_username("user123") == True
assert is_valid_username("12user") == False
assert is_valid_username("us er") == False
assert is_valid_username("abcd") == False
assert is_valid_username("validuser99") == True
assert is_valid_username("user 123") == False
assert is_valid_username("verylongusername123") == False

print("username validation logic successfully passing all ai-generated test cases.")
```

Output:

Task Description #2 (Even–Odd & Type Classification – Apply AI for Robust Input Handling)

- Task: Use AI to generate at least 3 assert test cases for a function classify_value(x) and implement it using conditional logic and loops.
- Requirements:
  - o If input is an integer, classify as "Even" or "Odd".
  - o If input is 0, return "Zero".

o If input is non-numeric, return "Invalid Input".

Example Assert Test Cases:

assert classify_value(8) == "Even"

assert classify_value(7) == "Odd"

assert classify_value("abc") == "Invalid Input" Expected

Output #2:

Function correctly classifying values and passing all test cases.

Code:

```python
def classify_value(x):
    '''
    classifies the input value.

    rules:
    - if input is integer → even or odd
    - if input is 0 → zero
    - if input is non-numeric → invalid input
    '''

    if isinstance(x, int):

        if x == 0:
            return "Zero"

        for _ in range(1):
            if x % 2 == 0:
                return "Even"
            else:
                return "Odd"

    return "Invalid Input"


# assert test cases
assert classify_value(8) == "Even"
assert classify_value(7) == "Odd"
assert classify_value(0) == "Zero"
assert classify_value("abc") == "Invalid Input"
assert classify_value(4.5) == "Invalid Input"
assert classify_value(-3) == "Odd"

print("function correctly classifying values and passing all test cases.")
```

Output:

```
PS C:\Users\rohit\OneDrive\Desktop\lat.py> & C:/Users/rohit/AppData/Local/Microsoft/WindowsApps/python3.11.exe c:/Users/rohit/OneDrive/Desktop/lat.py/day1.py
function correctly classifying values and passing all test cases.
PS C:\Users\rohit\OneDrive\Desktop\lat.py>
```

Task Description #3 (Palindrome Checker – Apply AI for String Normalization)

- Task: Use AI to generate at least 3 assert test cases for a

function is_palindrome(text) and implement the function.

- Requirements:

o Ignore case, spaces, and punctuation.

o Handle edge cases such as empty strings and

single characters.

Example Assert Test Cases:

assert is_palindrome("Madam") == True

assert is_palindrome("A man a plan a canal Panama") ==

True

assert is_palindrome("Python") == False Expected

Output #3:

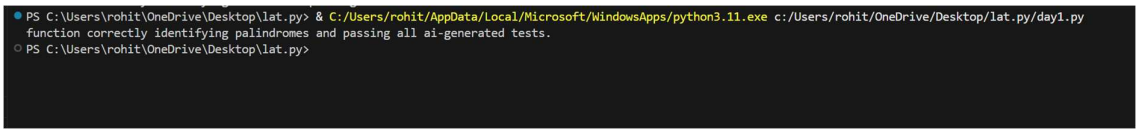
- Function correctly identifying palindromes and passing all

AI-generated tests.

Code:

```
def is_palindrome(text):
    '''
    checks whether the given text is a palindrome.

    rules:
    - ignore case, spaces, and punctuation
    - empty string is considered a palindrome
    - single character is also a palindrome
    '''

    number = ""
    for ch in text.lower():
        if ch.isalnum():
            number += ch
    return number == number[::-1]


# assert test cases
assert is_palindrome("Madam") == True
assert is_palindrome("A man a plan a canal Panama") == True
assert is_palindrome("Python") == False
assert is_palindrome("") == True
assert is_palindrome("a") == True
assert is_palindrome("No lemon, no melon!") == True

print("function correctly identifying palindromes and passing all ai-generated tests.")
```

Output:



Task Description #4 (BankAccount Class – Apply AI for

Object-Oriented Test-Driven Development)

- **Task: Ask AI to generate at least 3 assert-based test cases**

for a BankAccount class and then implement the class.

- **Methods:**

o deposit(amount)

o withdraw(amount)

o get_balance()

Example Assert Test Cases:

acc = BankAccount(1000)

acc.deposit(500)

assert acc.get_balance() == 1500

acc.withdraw(300)

assert acc.get_balance() == 1200

Expected Output #4:

- **Fully functional class that passes all AI-generated assertions.**

Code:

```python
class BankAccount:
    '''
    simple bank account class.
    methods:
    - deposit(amount)
    - withdraw(amount)
    - get_balance()
    '''

    def __init__(self, balance):
        self.balance = balance

    def deposit(self, amount):
        if amount > 0:
            self.balance += amount

    def withdraw(self, amount):
        if amount > 0 and amount <= self.balance:
            self.balance -= amount

    def get_balance(self):
        return self.balance


acc = BankAccount(1000)
acc.deposit(500)
assert acc.get_balance() == 1500

acc.withdraw(300)
assert acc.get_balance() == 1200

acc.withdraw(2000)
assert acc.get_balance() == 1200

acc.deposit(-100)
assert acc.get_balance() == 1200

print("fully functional class that passes all ai-generated assertions.")

```

Output:

```
PS C:\Users\rohit\OneDrive\Desktop\lat.py> & C:/Users/rohit/AppData/Local/Microsoft/WindowsApps/python3.11.exe c:/Users/rohit/OneDrive/Desktop/lat.py/day1.py
fully functional class that passes all ai-generated assertions.
PS C:\Users\rohit\OneDrive\Desktop\lat.py>
```

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Task Description #5 (Email ID Validation – Apply AI for Data Validation)

- Task: Use AI to generate at least 3 assert test cases for a function validate_email(email) and implement the function.
- Requirements:

o Must contain @ and .

o Must not start or end with special characters.

o Should handle invalid formats gracefully. Example Assert Test Cases:

assert validate_email("user@example.com") == True

assert validate_email("userexample.com") == False

assert validate_email("@gmail.com") == False Expected

Output #5:

- Email validation function passing all AI-generated test cases

and handling edge cases correctly.

Code:

```python
def validate_email(email):
    '''
    checks whether the given email is valid.
    rules:
    - must contain @ and .
    - should not start or end with special characters
    - should handle invalid formats safely
    '''

    if not isinstance(email, str):
        return False

    if "@" not in email or "." not in email:
        return False

    if not email[0].isalnum() or not email[-1].isalnum():
        return False

    parts = email.split("@")
    if len(parts) != 2:
        return False

    local, domain = parts

    if local == "" or domain == "":
        return False

    if "." not in domain:
        return False

    if domain.startswith(".") or domain.endswith("."):
        return False

    return True


# assert test cases
assert validate_email("user@example.com") == True
assert validate_email("userexample.com") == False
assert validate_email("@gmail.com") == False
assert validate_email("user@.com") == False
assert validate_email(".user@gmail.com") == False
assert validate_email("user@gmail.com.") == False
assert validate_email("user123@test.co") == True

print("email validation function passing all ai-generated test cases.")

```

Output:

```
PS C:\Users\rohit\OneDrive\Desktop\lat.py> & C:/Users/rohit/AppDat
a/Local/Microsoft/WindowsApps/python3.11.exe c:/Users/rohit/OneDri
ve/Desktop/lat.py/day1.py
email validation function passing all ai-generated test cases.
PS C:\Users\rohit\OneDrive\Desktop\lat.py>
```